THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 2: **Exploring Mixtures and Their Separation** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A mixture of potassium nitrate and sodium chloride is separated by: 1

(a) filtration **(b)** sublimation
(c) fractional crystallisation **(d)** distillation

2. A saturated solution of 140 g contains 40 g solute. The solubility per 100 g water is: 1

(a) 28.6 g **(b)** 100 g **(c)** 40 g **(d)** 35 g

3. Which of the following sets are all colloids? 1

(a) muddy water, chalk, flour **(b)** milk, fog, butter
(c) salt solution, air, brass **(d)** sugar solution, vinegar, tea

4. The most suitable method to separate a mixture of ethanol (b.p. 78 degrees Celsius) and water is: 1

(a) decantation **(b)** fractional distillation
(c) filtration **(d)** magnet

5. The solubility of a gas in water decreases when: 1

(a) temperature increases **(b)** pressure increases
(c) temperature decreases **(d)** water is cold

6. A pure compound differs from a mixture because it: 1

(a) has a fixed melting point **(b)** has variable composition
(c) is separated by filtration **(d)** has no formula

7. A student removes the cream from milk by centrifuging. The cream rises because: 1

(a) it is more dense **(b)** it is hot **(c)** it is solid **(d)** it is less dense

8. Which mixture shows the Tyndall effect and does not settle? 1

(a) milk **(b)** salt in water **(c)** sand in water **(d)** chalk in water

9. Which among the following is NOT a physical method of separation? 1

(a) electrolysis of water **(b)** sublimation
(c) evaporation **(d)** filtration

10. What type of mixture is smoke? 1

(a) true solution **(b)** colloid (solid in gas)
(c) pure **(d)** compound

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. How is the concentration of a dilute solution changed by adding the solvent? 2

12. Why does a solution of copper sulphate crystallise when it cools? 2

13. Why must the filter paper be wet with solvent before filtration? 2

14. Why can colloids not be separated by simple filtration? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain why fractional distillation is better than simple distillation for liquids with close boiling points. 3

16. Describe how to purify an impure sample of salt containing sand. 3

17. Explain why a mixture does not have a fixed melting point. 3

OR

Explain the separation of components in chromatography of ink using solubility and adsorption. 3

18. State three methods of separating a gas from a mixture of gases. 3

OR

Explain the process of obtaining common salt from sea water in India. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the preparation of a saturated solution of KNO3 at 30 degrees Celsius and the use of cooling to crystallise. 5

OR

Describe how a mixture of two miscible liquids such as ethanol and water is separated and explain why the first fraction is not pure ethanol. 5

20. Compare solutions, colloids and suspensions in a table with five properties and examples. 5

OR

Explain each term with an example: miscible, immiscible, solute, solvent, solubility, solution. 5

END OF QUESTION PAPER — Total: 40 Marks

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